

Information Retrieval

UNIT 1 – Foundations of Information Retrieval

● TOP PRIORITY QUESTIONS (Must Study)

Introduction to Information Retrieval

1. Define **Information Retrieval (IR)** and explain its **components** with diagram.
2. Explain the **challenges and applications of Information Retrieval systems**.

Document Indexing, Storage, and Compression

3. Explain **Inverted Index construction** with example.

(This is one of the MOST repeated questions)

4. Construct an **Inverted Index for a given document collection**.

(This numerical/problem question is VERY frequent)

Retrieval Models

5. Explain the **Boolean Retrieval Model** with suitable example and query processing.
6. Explain the **Vector Space Model**, including **TF-IDF and cosine similarity**.

(This is one of the MOST IMPORTANT questions)

7. Explain the **Probabilistic Retrieval Model** and relevance feedback.
8. Explain Term Weighting and Document Representation in IR.

Spelling Correction in IR Systems

9. Explain **Edit Distance (Levenshtein distance)** with example and its role in spelling correction.

(This numerical question appears frequently)

10. Explain techniques used for **Spelling Correction in IR systems**, including Soundex algorithm.

11. Challenges of spelling errors in queries and documents

Performance Evaluation

12. Define and explain **Precision, Recall, and F-measure** with example.

(This is compulsory exam question in most papers)

13. Explain the role of **Test Collections and Relevance Judgments** in evaluating IR systems.

ADDITIONAL PRACTICE QUESTIONS

Introduction to IR

14. Explain the **need and characteristics of Information Retrieval systems**.

Document Indexing

15. Explain **Tokenization and Stop Words** and their importance in IR.

16. Construct **Term-Document Incidence Matrix** and process Boolean queries.

Retrieval Models

17. Explain **TF-IDF weighting scheme** and its importance in ranking.

18. What is **Relevance Feedback**? Explain its role in retrieval models.

Spelling Correction

19. Explain **Soundex Algorithm** for phonetic matching.

20. Explain **n-gram indexing (2-gram, 3-gram)** used in tolerant retrieval.(Problems question too)

Performance Evaluation

21. Explain **Average Precision** and its importance in IR evaluation.

UNIT 2 – Advanced Topics in Information Retrieval

● TOP PRIORITY QUESTIONS (Must Study)

Text Categorization and Filtering

1. Explain the **Naive Bayes algorithm for text classification** with working and assumptions.
2. Explain **Support Vector Machines (SVM) for text classification**.

Text Clustering

3. Explain **K-Means clustering algorithm** with example.
4. Explain **Hierarchical clustering** and differentiate it from K-Means clustering.
5. Explain how clustering is used for **query expansion and result grouping** in Information Retrieval.

Web Information Retrieval

6. Explain the **architecture of a Web Search Engine**, including crawling, indexing, and query processing.
7. Explain **Web Crawling and Indexing process**.

Learning to Rank

8. What is **Learning to Rank**? Explain **supervised learning approaches** such as RankSVM and RankBoost.

Link Analysis (MOST IMPORTANT CHAPTER)

9. Define and explain the **PageRank algorithm** with example.

(This is one of the MOST repeated questions)

10. Explain the **HITS algorithm**, including hubs and authorities.
11. Differentiate between **PageRank and HITS algorithm**.
12. Explain Hubs and Authorities concept in detail.

ADDITIONAL PRACTICE QUESTIONS

Text Categorization

13. Explain **Feature Selection and Dimensionality Reduction** in text classification.

Text Clustering

14. Explain **evaluation of clustering results**.

Web Information Retrieval

15. Explain **challenges in Web Information Retrieval**.
16. What is a **snippet**? Explain its importance in Information Retrieval.

Learning to Rank

17. Explain **Pairwise and Listwise Learning to Rank approaches**.

Link Analysis

18. Explain role of **Link Analysis in Information Retrieval systems**.

Ranking Concept

19. Explain how **TF-IDF is used in ranking web pages**.

UNIT 3 – Advanced Topics in Information Retrieval

● TOP PRIORITY QUESTIONS (Must Study)

Crawling and Near-Duplicate Detection

1. Explain **Breadth-First and Depth-First Web Crawling techniques**.
2. Define **Near-Duplicate Page Detection** and explain its significance and challenges.

Text Summarization, Question Answering and Recommender Systems

3. Explain **Extractive and Abstractive Text Summarization** with differences.
4. Discuss challenges in **Abstractive Text Summarization**.
5. What is a **Question Answering System**? Explain its working and need.
6. Explain approaches used for finding precise answers in Question Answering systems.
7. Define and explain **Collaborative Filtering**.
8. Define and explain **Content-Based Filtering** and differentiate it from Collaborative Filtering.

(This is one of the MOST repeated questions)

Cross-Lingual Retrieval

9. Define **Cross-Lingual Information Retrieval** and explain its process and challenges.

10. Explain different approaches to **Machine Translation**:

- Rule-based
- Statistical
- Neural Machine Translation

Evaluation Techniques

11. Explain role of **Test Collections and Benchmarking datasets** in evaluating IR systems.

ADDITIONAL PRACTICE QUESTIONS

Crawling

12. Explain **Focused Crawling** and handling dynamic web content.

Text Summarization, Question Answering and Recommender Systems

13. Explain techniques used in **Extractive Text Summarization**.

14. Explain challenges associated with Question Answering systems.

15. Explain **User-based and Item-based Collaborative Filtering**.

Cross-Lingual Retrieval

16. Explain **Multilingual document representation and query translation**.

Evaluation

17. Explain **User-based evaluation methods** (User studies and surveys).

18. Explain **Online evaluation methods: A/B testing and interleaving experiments**.